Minor Project Synopsis Report

Disaster Allocation System

Project Category: Deep-tech

Projexa Team Id- 26E2033

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

in

CSE

to

K.R Mangalam University

By

Ananya Sharma (2401010056)
Diksha kumari (2401010055)
Krishika Sinha (2401010113)
Taniya (2401010033)
Shikha (2401010063)

Under the supervision of

Mr. Nandan Kumar Mishra

Department of Computer Science and Engineering

School of Engineering and Technology

K.R Mangalam University, Gurugram- 122001, India

January 2026

DECLARATION

We hereby declare that this project report is our original work and has not been submitted elsewhere. The information presented here is true and correct to the best of our knowledge.

Shikha (2401010063)
Ananya Sharma (2401010056)
Diksha kumari (2401010055)
Krishika Sinha (2401010113)
Taniya (2401010033)

INDEX

S No	Content	Page No
1.	Abstract	4
2.	Introduction	5
3.	Motivation	6
4.	Literature Review	7-9
5.	Gap Analysis	10
6.	Problem Statement	11
7.	Objectives	12
8.	Tools/platform Used	13-14
9.	Methodology	15-17
10.	Experimental Setup	18-21
11.	Results And Discussion	22
12.	Conclusion	23
13.	Future Work	24
14.	References	25
15.	Annexure I: Plagiarism Declaration Certificate	26
16.	Annexure II: Plagiarism Report	27-31

ABSTRACT

Disaster situations such as floods, earthquakes, fires, and other natural or man-made calamities often result in severe loss of life and property. One of the major challenges during such events is the **inefficient and delayed allocation of available resources** like food supplies, medical aid, rescue teams, and shelter facilities. Lack of proper coordination and real-time information can lead to unequal distribution, wastage of resources, and delayed response in critical areas.

The **Disaster Allocation System** aims to address these challenges by providing a **centralized and intelligent platform** for managing disaster-related information and resource distribution. The system is designed to collect, organize, and process disaster data in order to assist authorities in making **faster and more informed decisions**. By maintaining structured records of disaster incidents, available resources, and allocation details, the system ensures transparency and efficiency in emergency response operations.

The project is developed using modern backend technologies, enabling secure handling of data and modular system design. It focuses on scalability, so that the system can be extended in the future to include real-time analytics, geographic mapping, and automated decision support. The proposed solution helps in minimizing response time, avoiding resource duplication, and ensuring that aid reaches the most affected areas promptly.

Overall, the Disaster Allocation System contributes to improved disaster management by enhancing coordination, optimizing resource utilization, and supporting timely relief operations, thereby reducing the overall impact of disasters on society.

1. INTRODUCTION

Rapid urbanization, climate change, unplanned infrastructure growth, and population expansion have significantly increased the frequency and impact of disasters across the world. Natural disasters such as floods, earthquakes, cyclones, and landslides, along with man-made emergencies like industrial accidents and fires, pose serious threats to human life and economic stability. In developing countries like India, these challenges are further intensified due to inadequate disaster preparedness, limited resources, and poor coordination among relief agencies.

One of the major issues observed during disaster situations is the **inefficient allocation and management of relief resources**. Even when sufficient aid is available, the absence of a structured system often results in delayed response, unequal distribution, and mismanagement of essential supplies such as food, medical assistance, shelter, and rescue services. This lack of coordination can worsen the situation, leading to avoidable losses and prolonged recovery periods.

The COVID-19 pandemic further exposed the vulnerabilities of existing emergency response systems. Lockdowns, supply chain disruptions, and sudden surges in demand highlighted the need for a reliable and centralized mechanism to manage emergency resources efficiently. Similar challenges are observed during large-scale disasters, where manual record-keeping and unorganized communication fail to meet the urgency of the situation.

In such scenarios, a technology-driven solution becomes essential. A **Disaster Allocation System** provides a centralized platform that helps authorities monitor disaster events, track available resources, and ensure timely and fair distribution to affected areas. By enabling structured data management and improved coordination, the system plays a crucial role in strengthening disaster response mechanisms and reducing the overall impact of emergencies on society.

2. MOTIVATION

In recent years, the world has witnessed a rapid increase in the frequency and severity of disasters due to climate change, uncontrolled urbanization, population growth, and inadequate planning. Natural calamities such as floods, earthquakes, cyclones, and landslides, along with man-made disasters like industrial accidents and fires, have caused massive loss of life and property. In countries like India, these situations are often aggravated by poor coordination among authorities and inefficient distribution of relief resources.

Although disaster management authorities make significant efforts to provide aid, the lack of a centralized and well-structured allocation system results in delays, duplication of resources, and unequal distribution of relief. In many cases, some regions receive excess supplies while others suffer due to shortages of essential necessities such as food, medical aid, shelter, and rescue equipment. These inefficiencies highlight the urgent need for a system that can streamline disaster response and ensure optimal utilization of available resources.

The COVID-19 pandemic further exposed the shortcomings of traditional emergency management systems. Sudden spikes in demand, logistical challenges, and lack of real-time information made it extremely difficult to manage and allocate resources effectively. Similar challenges are observed during large-scale disaster events, where manual processes and fragmented data lead to poor decision-making under time-sensitive conditions.

The motivation behind this project is to design and develop a **Disaster Allocation System** that enhances the efficiency, transparency, and speed of resource distribution during emergency situations. By providing a centralized platform to manage disaster information, track available resources, and support informed allocation decisions, the system aims to minimize response delays and reduce the overall impact of disasters. This project seeks to contribute toward a more reliable and technology-driven disaster management framework that can support authorities in saving lives and restoring normalcy more effectively.

3. LITERATURE REVIEW

DISASTER MANAGEMENT AND RESOURCE ALLOCATION SYSTEMS:

Several studies highlight the critical role of efficient resource allocation in disaster management. Research emphasizes that improper planning and delayed distribution of relief materials significantly increase casualties and economic losses during disasters. Traditional disaster response systems largely depend on manual coordination among multiple agencies, which often leads to miscommunication, duplication of efforts, and uneven distribution of essential resources. Studies suggest that centralized disaster management platforms can improve coordination by enabling authorities to monitor disaster severity, available resources, and response status in real time, thereby improving decision-making during emergencies.

ICT-BASED DISASTER RESPONSE SYSTEMS:

The integration of Information and Communication Technology (ICT) in disaster management has been widely explored to improve emergency response efficiency. Several researchers propose web-based and cloud-based disaster management systems that allow authorities to collect, store, and analyze disaster-related data. These systems enable faster dissemination of information among stakeholders such as government agencies, NGOs, and rescue teams. However, many existing ICT-based systems focus primarily on data reporting and lack effective mechanisms for automated or optimized resource allocation, which limits their practical effectiveness during large-scale disasters

RESOURCE TRACKING AND ALLOCATION USING DATABASE SYSTEMS:

Various studies have explored database-driven approaches for managing disaster resources such as food supplies, medical aid, rescue equipment, and shelters. These systems maintain structured records of resources, their locations, and availability status. While such approaches improve transparency and record-keeping, they often rely on static data and manual updates, making them less suitable for rapidly changing disaster scenarios. Researchers emphasize the need for dynamic systems that can

update resource availability in real time and support quick allocation decisions based on urgency and demand.

DECISION SUPPORT SYSTEMS FOR DISASTER MANAGEMENT;

Decision Support Systems (DSS) have been proposed to assist authorities in evaluating disaster severity and prioritizing resource distribution. These systems use predefined rules, historical data, and analytical models to suggest optimal allocation strategies. Studies show that DSS can significantly reduce response time and improve resource utilization. However, many existing systems are complex, expensive to implement, and require extensive training, which restricts their adoption in developing countries.

WEB-BASED DISASTER MANAGEMENT PLATFORMS;

Recent research highlights the growing adoption of web-based disaster management platforms due to their accessibility and scalability. These platforms allow multiple users to access the system simultaneously, update disaster information, and track relief operations from different locations. While such systems improve collaboration, most lack role-based access control and proper authorization mechanisms, raising concerns regarding data security and misuse. This creates a need for a secure, role-driven disaster allocation system that balances accessibility with data integrity.

Table: Literature Review on Disaster Management and Resource Allocation Systems

S. No.	Title / Area	Methodology Used	Key Findings	Limitations Identified
1	Disaster Management and Resource Allocation	Manual coordination and centralized planning	Highlighted the importance of timely resource allocation to reduce casualties	Manual processes lead to delays and miscommunication
2	ICT-Based Disaster Response Systems	Web-based information systems	Improved communication between disaster response agencies	Lack of automated resource allocation mechanisms
3	Database-Driven Disaster Resource Management	Relational databases for resource tracking	Enhanced transparency and record maintenance	Static data; not suitable for real-time disaster scenarios
4	Decision Support Systems for Disaster Management	Rule-based decision support models	Assisted authorities in prioritizing resource distribution	Complex implementation and high cost
5	Cloud-Based Disaster Management Platforms	Cloud storage and online dashboards	Enabled remote access and multi-agency collaboration	Security and access control concerns
6	Emergency Response Coordination Systems	Centralized control systems	Reduced response time during emergencies	Poor scalability during large-scale disasters
7	Real-Time Disaster Monitoring Systems	Sensor and data-driven monitoring	Provided faster situational awareness	High dependency on infrastructure availability
8	Web-Based Relief Distribution Systems	Online portals for relief tracking	Improved accountability and tracking of relief materials	Limited user role management and authorization
9	Automated Disaster Resource Allocation Models	Optimization and scheduling algorithms	Improved efficiency in resource utilization	Difficult to implement in real-world dynamic conditions
10	Integrated Disaster Management Systems	Multi-module system architecture	Combined disaster reporting, monitoring, and response	Lack of modularity and flexibility

4. GAP ANALYSIS

From the extensive research conducted in the field of disaster management, it is evident that technology has been increasingly used to improve disaster response through ICT-based systems, web platforms, database management, and decision support tools. Many existing studies focus on disaster reporting, early warning systems, and post-disaster assessment, which help authorities understand the severity and impact of disasters. While these systems enhance information sharing and documentation, they often function in isolation and lack integrated mechanisms for efficient resource allocation.

Most current disaster management solutions emphasize data collection and communication rather than real-time, intelligent allocation of relief resources such as food, medical aid, rescue teams, and shelters. In many cases, resource allocation decisions are still carried out manually, leading to delays, duplication of efforts, unequal distribution, and wastage of critical supplies. Additionally, several decision support systems proposed in research are complex, costly, and difficult to deploy, making them unsuitable for developing countries with limited infrastructure and technical expertise.

Furthermore, existing systems rarely provide a centralized platform that combines disaster incident tracking, resource availability monitoring, and allocation records in a structured and transparent manner. The absence of such an integrated approach reduces coordination among authorities and relief agencies, especially during large-scale emergencies where timely response is crucial.

The proposed Disaster Allocation System addresses these gaps by offering a centralized, scalable, and structured platform for managing disaster-related data and resource distribution. By focusing on efficient allocation, transparency, and future extensibility, the system aims to overcome the limitations of existing solutions and provide a practical, technology-driven approach to improve disaster response and relief operations.

5. PROBLEM STATEMENT

During disaster situations such as floods, earthquakes, cyclones, fires, and pandemics, timely and efficient allocation of relief resources is critical for saving lives and minimizing damage. However, existing disaster management practices often rely on manual coordination, fragmented data, and unstructured communication among multiple agencies. This results in delays in response, unequal distribution of resources, duplication of efforts, and wastage of essential supplies such as food, medical aid, rescue teams, and shelter facilities.

In many cases, authorities lack a centralized system to track disaster incidents, monitor real-time availability of resources, and record allocation details. The absence of an integrated and transparent platform makes it difficult to prioritize affected areas based on severity and urgency, leading to inefficient decision-making during time-sensitive emergencies. These challenges are further intensified in large-scale disasters where rapid response and coordination are crucial.

Therefore, the problem lies in designing a centralized and efficient Disaster Allocation System that can systematically manage disaster information, track available resources, and support timely and fair allocation of relief resources. Such a system should improve coordination among authorities, reduce response time, and ensure that aid reaches the most affected regions effectively.

6. OBJECTIVES

Sample Objectives

1. **To design and develop a centralized Disaster Allocation System** that efficiently manages disaster-related information and coordinates the distribution of relief resources during emergency situations.
2. **To create a structured database** for maintaining real-time records of disaster incidents, available resources such as food, medical aid, rescue teams, and shelters, and their allocation status.
3. **To improve decision-making during disasters** by enabling authorities to prioritize affected areas based on severity, urgency, and resource availability.
4. **To ensure transparency and accountability** in the allocation process by maintaining detailed allocation logs and minimizing duplication or misuse of resources.
5. **To develop a scalable and secure system architecture** that can be extended in the future to support real-time analytics, geographic mapping, and automated decision-support features.

The objective of this project is to bridge the gaps identified in existing disaster management systems by providing a unified and efficient platform for disaster resource allocation. The proposed system aims to reduce response time, optimize resource utilization, and enhance coordination among authorities, thereby supporting effective disaster relief operations and minimizing the overall impact of disasters on society.

7. Tools/Technologies Used

For the development of the Disaster Allocation System, a combination of modern frontend, backend, and database technologies has been used to ensure efficient data management, user-friendly interaction, system scalability, and reliability during emergency situations. Each technology has been carefully selected based on its suitability for handling disaster-related information and resource allocation processes.

FRONTEND TECHNOLOGIES

HTML (HyperText Markup Language)

HTML is used to design the basic structure of the web application. It provides the layout for different modules such as disaster reporting, resource management, and allocation dashboards. HTML enables the creation of forms, tables, and reports required for systematic data entry and display.

CSS (Cascading Style Sheets)

CSS is used to enhance the visual appearance of the web application. It helps in designing a responsive, clean, and user-friendly interface by controlling layout, colors, fonts, and spacing. A well-designed interface is essential for quick access to information during disaster situations.

JavaScript

JavaScript is used to add interactivity and dynamic behavior to the frontend. It enables real-time form validation, dynamic content updates, and smooth user interaction without frequent page reloads. JavaScript improves system responsiveness and enhances the overall user experience.

BACKEND TECHNOLOGY

The backend of the Disaster Allocation System is responsible for processing user requests, managing disaster and resource data, and enforcing system logic. It handles authentication, authorization, and communication between the frontend and the database. The backend ensures secure, reliable, and efficient data flow, which is critical during emergency response operations.

DATABASE MANAGEMENT SYSTEM: MONGODB

MongoDB is used as the Database Management System (DBMS) for the Disaster Allocation System. It is a NoSQL, document-oriented database that stores data in JSON-like documents, making it suitable for managing dynamic and large-scale disaster-related data.
MongoDB provides high performance, flexibility, and scalability, which are essential for handling rapidly changing information such as disaster incidents, resource availability, and allocation records. Its schema-less structure allows easy modification and future expansion of the system. Additionally, MongoDB supports high availability and fast data retrieval, ensuring uninterrupted system operation during critical situations.

REASONS FOR SELECTING THESE TECHNOLOGIES

1. **Efficient Data Handling** – Supports structured and dynamic disaster data management.
2. **Scalability** – Allows the system to grow with increasing data and users.
3. **User-Friendly Interface** – Enables quick access and ease of use during emergencies.
4. **Fast Processing and Response** – Ensures timely allocation of resources.
5. **Secure and Reliable** – Protects sensitive disaster and allocation data.
6. **Future Extensibility** – Can be enhanced with analytics, mapping, and decision support features.

8. METHODOLOGY

1. DISASTER DATA COLLECTION MODULE

This module is responsible for registering and monitoring disaster incidents. Information such as disaster type, affected location, severity level, date, and description is collected through the system interface. The collected data is stored in the database in a structured format, allowing authorities to track multiple disaster events simultaneously.

This module ensures that disaster information is consistently updated, enabling authorities to analyze the situation accurately and respond promptly.

---

2. RESOURCE MANAGEMENT MODULE

The resource management module maintains detailed records of available relief resources such as food supplies, medical aid, rescue teams, transportation facilities, and shelters. Each resource entry includes information regarding quantity, location, availability status, and last updated time.

This module helps in preventing duplication of resources and ensures that authorities have real-time visibility of available aid, which is critical during large-scale disaster situations.

---

3. RESOURCE ALLOCATION MODULE

The resource allocation module is the core component of the system. It processes disaster severity and resource availability data to allocate resources efficiently to affected areas. Allocation decisions are made based on predefined criteria such as urgency, disaster impact, and resource availability.

Once resources are allocated, the allocation details are recorded in the database along with timestamps to ensure transparency and accountability. This systematic allocation process reduces delays and ensures fair distribution of relief materials.

---

4. MONITORING AND TRACKING MODULE

This module continuously tracks the status of allocated resources and disaster response activities. It allows authorities to monitor whether allocated resources have been dispatched, delivered, or are still pending. Any updates or changes are logged in the system for future reference.

The monitoring feature improves coordination among agencies and helps identify bottlenecks in relief operations.

---

5. USER MANAGEMENT AND ACCESS CONTROL MODULE

The system implements role-based access control to ensure data security and integrity. Different users such as administrators, disaster management officials, and field coordinators are granted access based on their roles and responsibilities.

This module prevents unauthorized access and ensures that sensitive disaster and allocation data is handled securely.

---

6. REPORTING AND LOGGING MODULE

This module generates structured reports related to disaster incidents, resource utilization, and allocation history. Logs are maintained for every critical operation, providing a transparent record of decisions and actions taken during disaster response.

These reports assist authorities in evaluating system performance and planning future disaster preparedness strategies.

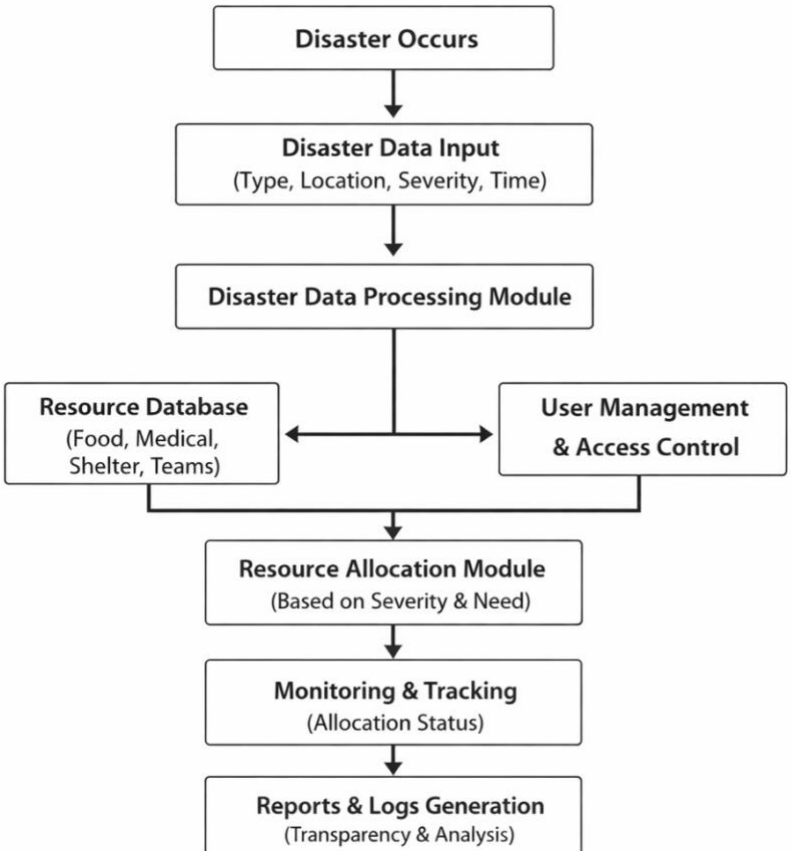

Figure 2: Architectural Diagram

This system is designed to manage disaster situations efficiently by processing disaster-related data and allocating resources accordingly. It begins with inputting disaster details such as type, location, severity, and time. The system then processes this data to determine priorities and interacts with a resource database (food, medical supplies, shelters, and teams) and a user management module for secure access. Based on the severity and need, the resource allocation module distributes resources, which are tracked in real-time by the monitoring module. Finally, reports and logs are generated for transparency, analysis, and future planning. The system can be implemented in Python using various modules for data handling, tracking, and reporting.

9.Experimental Setup

The Disaster Allocation System was developed and tested. The frontend was built using HTML, CSS, and JavaScript, while the backend was implemented using FastAPI in Python. MongoDB was used as the database for storing disaster, resource, and allocation data.

The system was tested by performing various operations such as adding disaster data, managing resources, and allocating resources through the frontend interface. API requests were handled by the backend and the results were stored and retrieved from the MongoDB database.

The system also includes role-based access control, where different users such as administrators and volunteers can log in and access functionalities based on their roles.

The following screenshots demonstrate the working of the complete system.

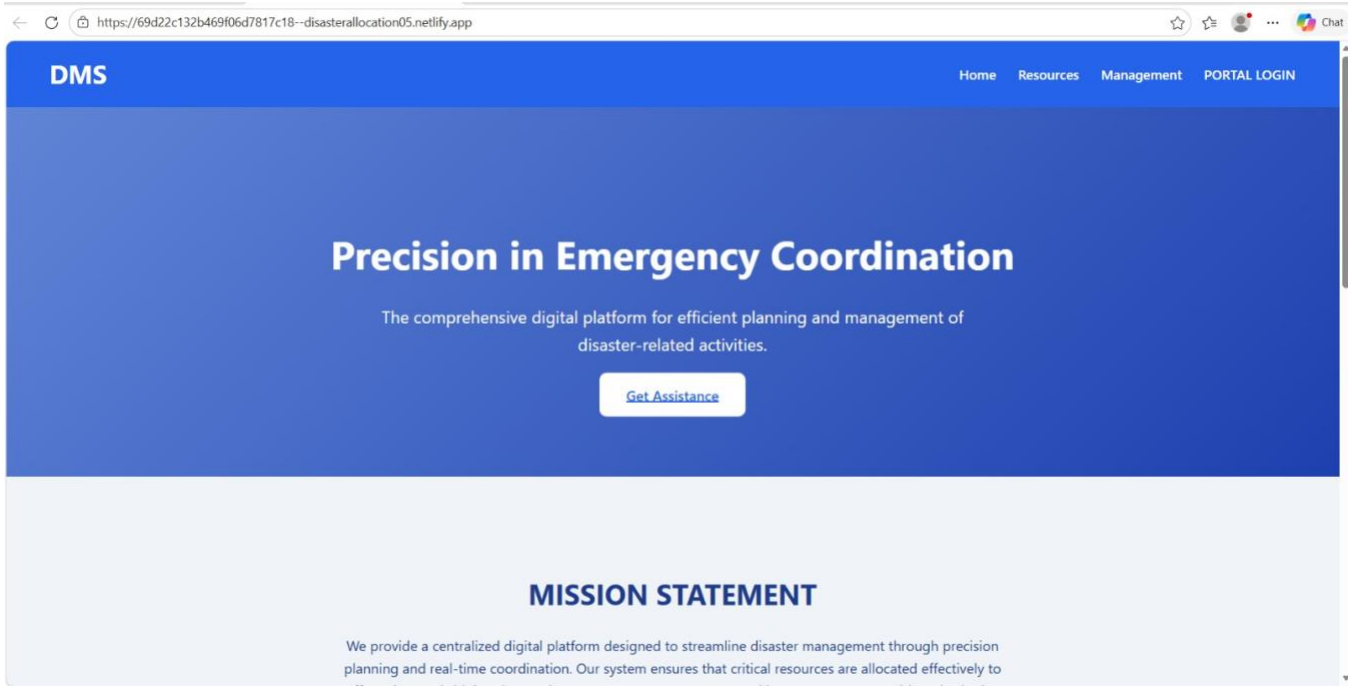

Figure 1: Frontend interface of the Disaster Allocation System

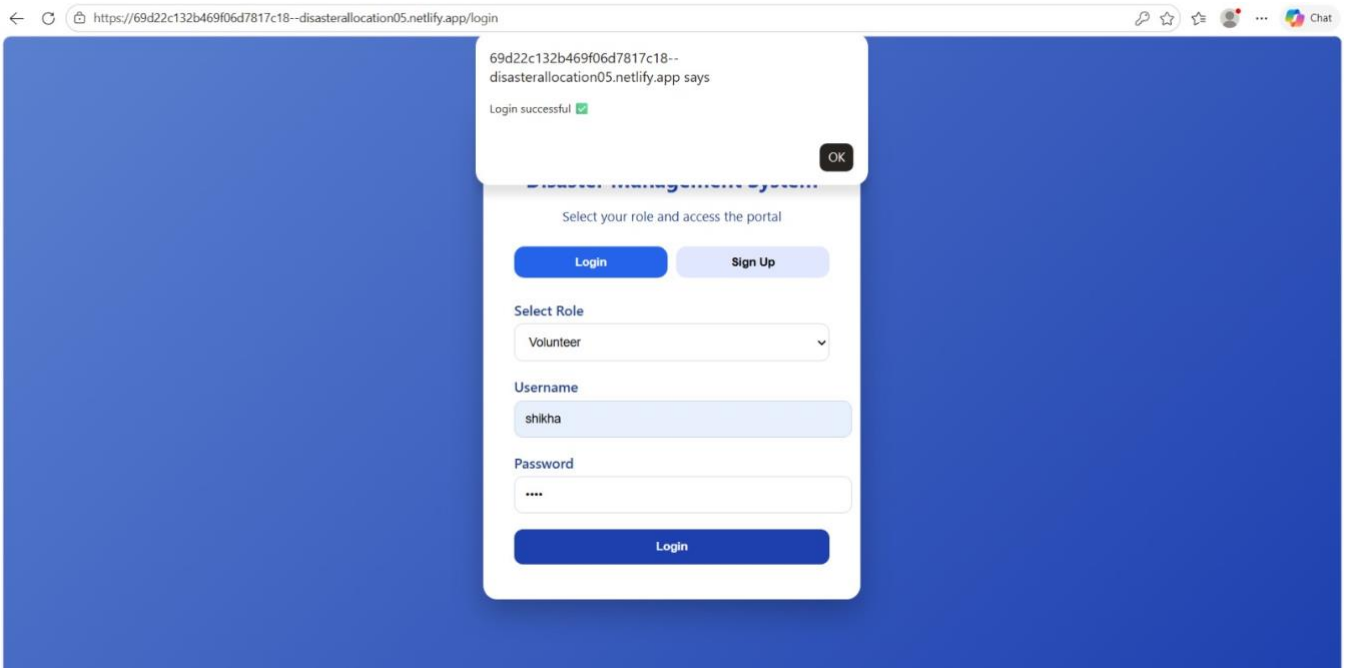

Figure 2: Volunteer login interface of the system.

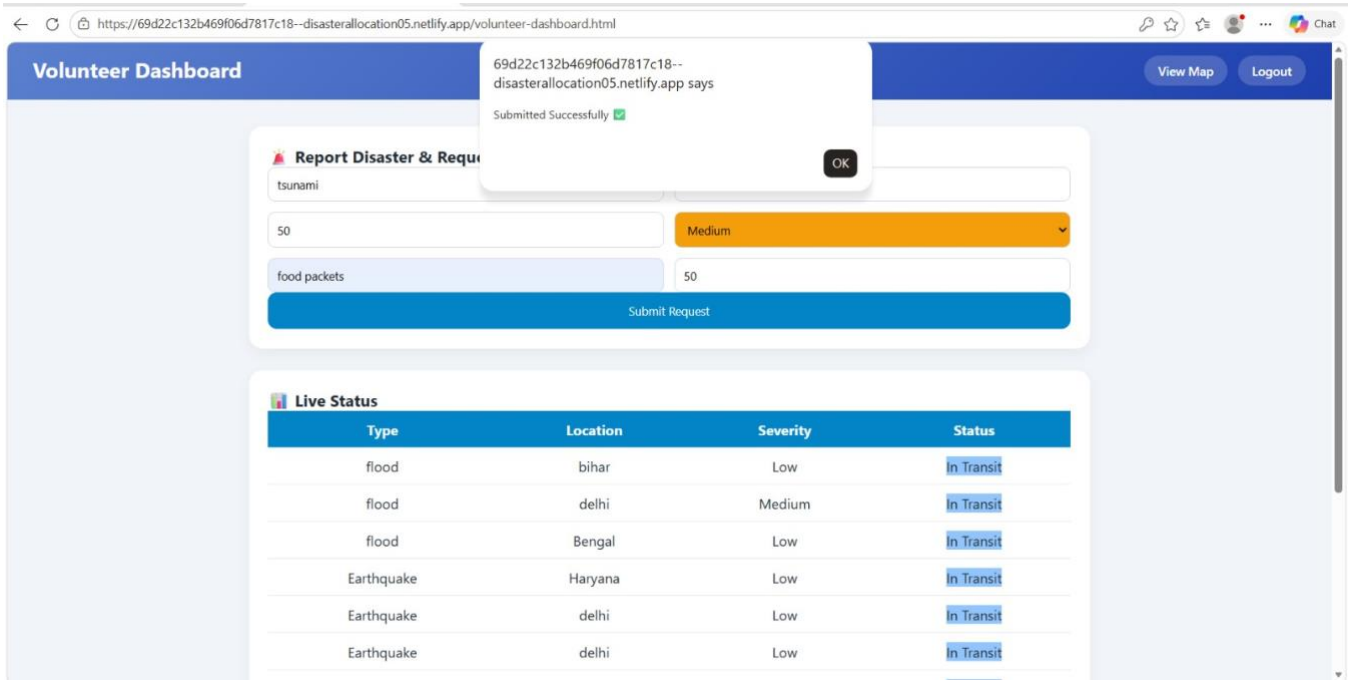

Figure 3: Adding disaster data through frontend interface

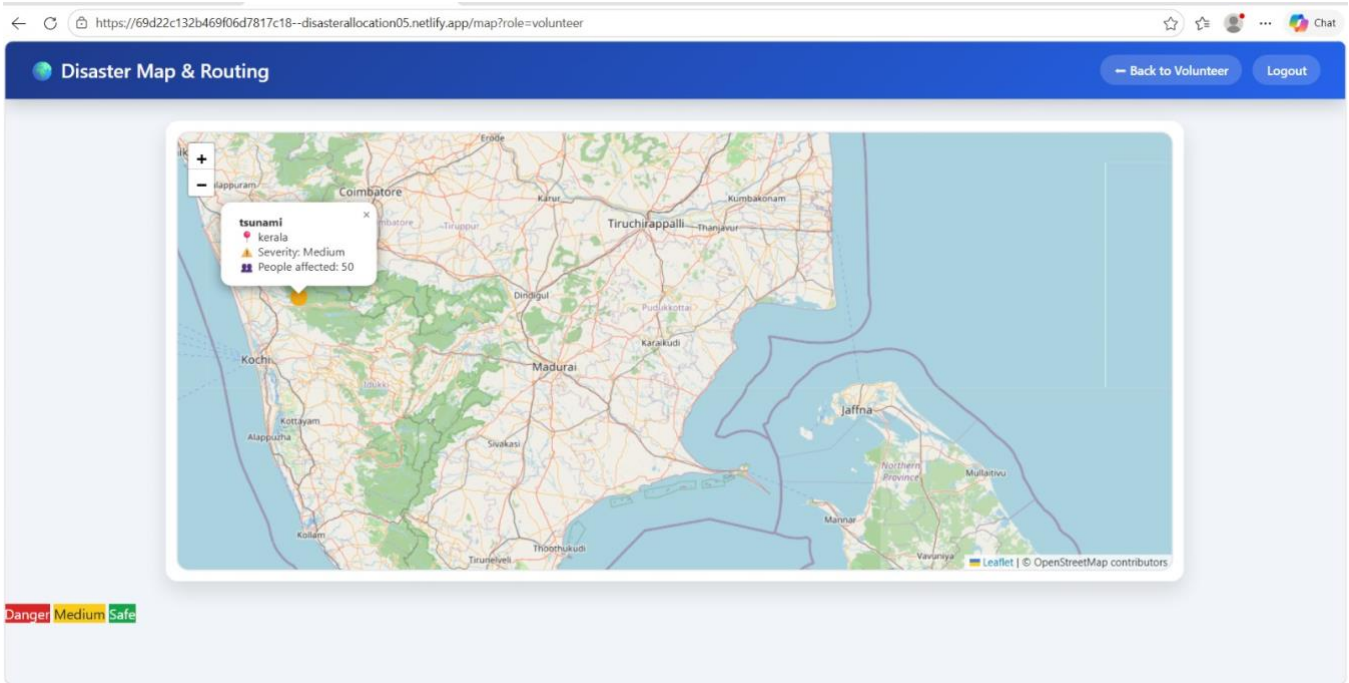

Figure 4: Map-based visualization showing disaster locations and resource requirements

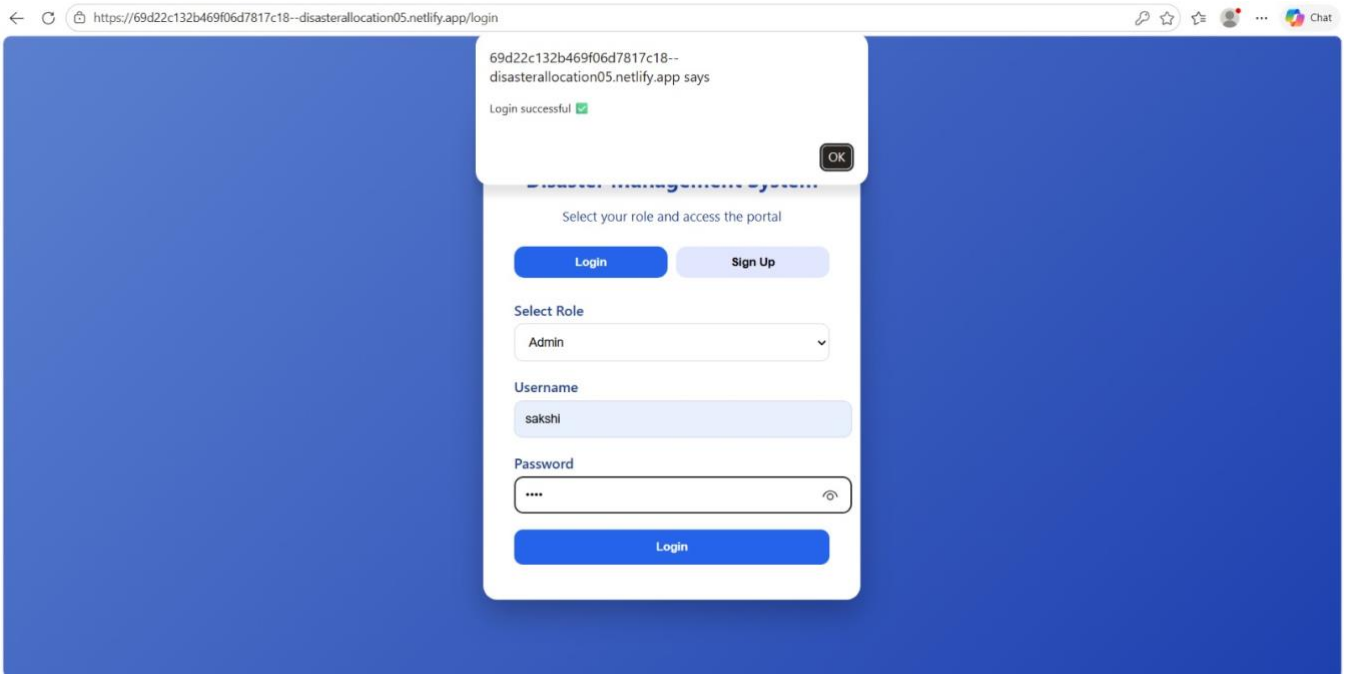

Figure 5: Admin login interface with access control features

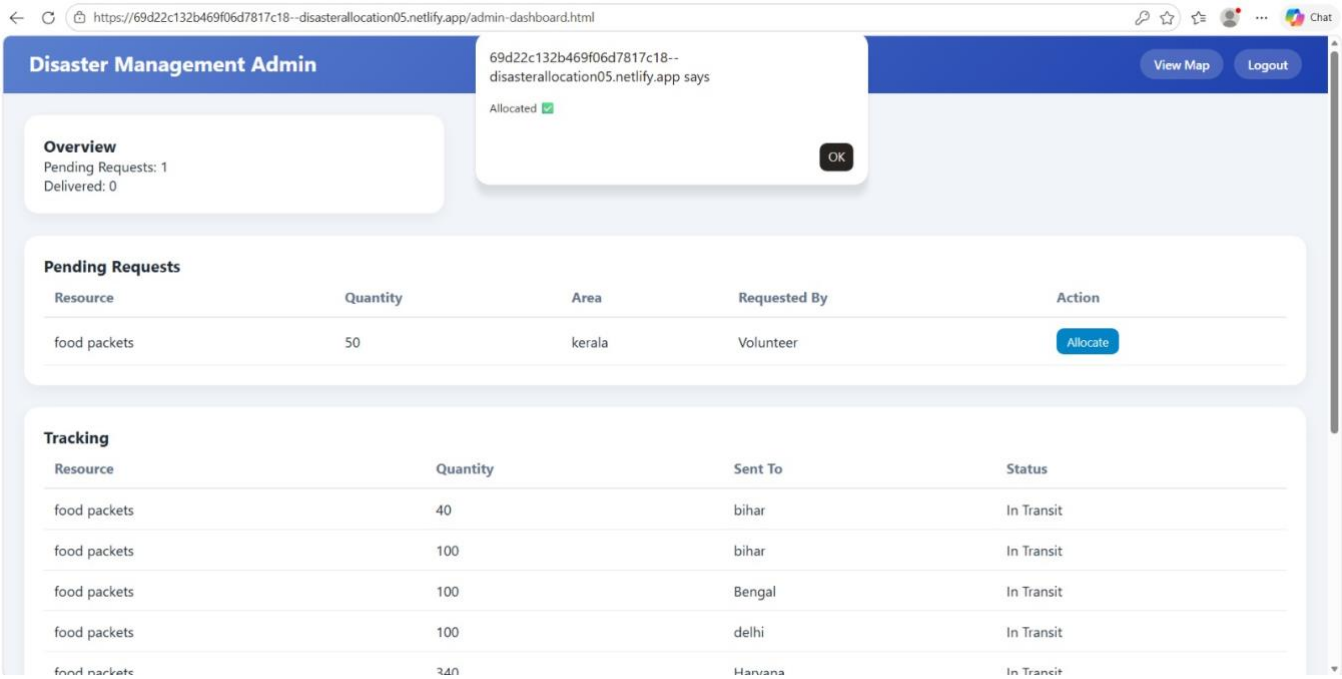

Figure 6: Resource allocation result based on disaster severity

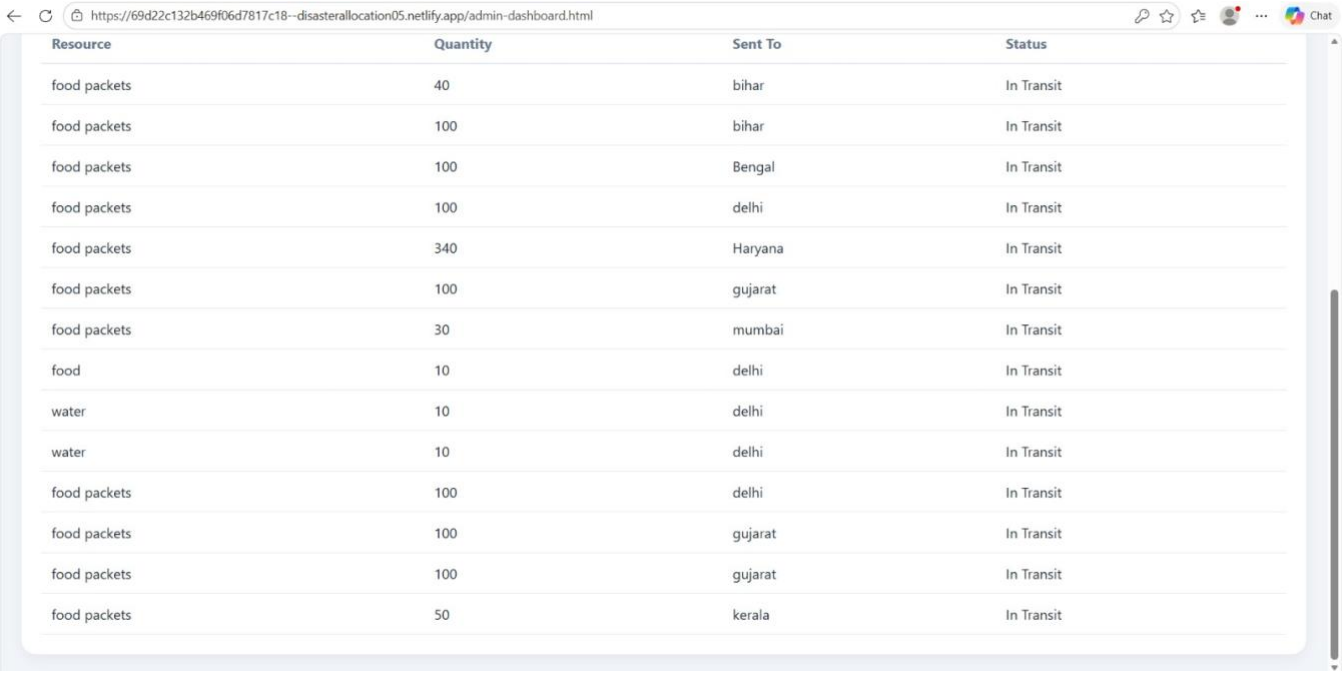

Figure 7: Resource status showing “In Transit” during delivery process

10.RESULTS AND DISCUSSION

The Disaster Allocation System was successfully implemented and tested.

Results:
- Disaster information is stored and retrieved correctly
- Resource management module works efficiently
- Allocation records are maintained properly
- APIs return correct responses during testing

Discussion:
The system improves coordination and ensures structured handling of data.
It reduces manual effort and helps in faster decision-making.

Limitations:
- Frontend and backend integration is partially completed
- No real-time data updates yet

11.CONCLUSION

The Disaster Allocation System has been successfully designed and developed to address the challenges faced during disaster situations, particularly in the efficient management and allocation of resources. The system provides a centralized platform for storing disaster-related data, tracking available resources, and maintaining allocation records in a structured manner.

By replacing manual processes with a digital system, it helps reduce delays, minimize resource wastage, and improve coordination among authorities. The project ensures transparency and better decision-making by maintaining proper logs and records of all operations.

Overall, the system contributes to improving disaster response efficiency and supports timely distribution of resources to affected areas. Although the current implementation focuses mainly on backend functionality, it provides a strong foundation for building a complete and scalable disaster management solution in the future.

12.FUTURE WORK

The Disaster Allocation System has been successfully developed with an integrated frontend and backend, providing a complete platform for managing disaster data and resource allocation. However, the system can be further enhanced with advanced features to improve its real-world applicability.

In the future, a real-time tracking system can be implemented to monitor the movement and delivery status of allocated resources such as food, medical supplies, and rescue teams. This will ensure that resources reach the affected areas on time and improve transparency in the distribution process.

Additionally, machine learning techniques can be incorporated to predict disaster severity and automatically suggest optimal resource allocation strategies based on historical data.

These enhancements will make the system more intelligent, scalable, and capable of handling complex real-world disaster management scenarios.

13.REFERENCES

- Su, Z., Zhang, G., Liu, Y., Yue, F., & Jiang, J. (2016). Multiple emergency resource allocation for concurrent incidents in natural disasters. *International Journal of Disaster Risk Reduction*, 16, 145–158. https://doi.org/10.1016/j.ijdrr.2016.01.007

- Altay, N. (2012). Capability-based resource allocation for effective disaster response. *IMA Journal of Management Mathematics*, 24(2), 253–273. https://doi.org/10.1093/imaman/dpr017

- Rye, S., & Aktas, E. (2022). A multi-attribute decision support system for allocation of humanitarian cluster resources. *Sustainability*, 14(20), 13423. https://doi.org/10.3390/su142013423

- Zhou, Y., Chen, X., & Huang, Z. (2023). Location–allocation problem in emergency logistics considering lateral transshipment strategy. *Computers & Industrial Engineering*, 180, 109365. https://doi.org/10.1016/j.cie.2023.109365

- Li, X., Sun, P., & Wang, H. (2025). Strategic allocation of medical resources in disaster response: Multi-objective dynamic optimization. *Computers & Industrial Engineering*, 190, 108551. https://doi.org/10.1016/j.cie.2025.108551

- Moghaddam, M. H., & Torabi, S. A. (2022). Multi-period equitable and efficient disaster relief allocation under uncertainty. *Annals of Operations Research*, 320, 531–558. https://doi.org/10.1007/s10479-022-04815-7

- Zhang, Y., & Huang, M. (2013). Resource redistribution under lack of information in disaster scenarios. *arXiv preprint arXiv:1312.6339*. https://arxiv.org/abs/1312.6339

- Zhao, Q., & Wang, L. (2021). Game-theoretic models for resource allocation during floods. *arXiv preprint arXiv:2112.01439*. https://arxiv.org/abs/2112.01439

14.ANNEXURE I:

Plagiarism Declaration Certificate

Title of Work: Disaster Allocation System

Submitted By: ByteX

Institution: K. R. Mangalam University, Gurugram

Department: SOET

We hereby declare that the work entitled **" Disaster Allocation System "** submitted for academic evaluation and research purposes is our original work. We confirm that:

- We have acknowledged and properly cited all sources, references, and data included in this work.
- This work does not contain any material previously published, written, or prepared by another person, except where due acknowledgment has been made.
- We understand that plagiarism is an academic offense and a violation of research ethics. Any breach of this declaration may lead to disciplinary action as per university policies.
- We have used appropriate referencing techniques and maintained academic integrity throughout this work.

We affirm that the submitted work has normal plagiarism less than 10% and free from AI content.

Signature of Student: _____

Date: _____

Plagiarismly

Plagiarized Content 2%

2% Exact 0% Partial

98% Unique Content

3854 Words

31469 Characters

Primary Sources

1	https://giftsandentertainment.roche.c... PSE/OSC Tech At CSE Expo Vegas: What You Need To ... Rochehttps://giftsandentertainment.roche.com › NewsRochehttps://giftsandentertainment.roche.com › News	0%
2	https://www.linkedin.com/pulse/unloc... Unlocking Potential: How Australia's Autism Strategy ... LinkedIn · Sandeep Bhasin, Ph.D.10+ reactions · 11 months agoLinkedIn · Sandeep Bhasin, Ph.D.10+ reactions · 11 months...	0%
3	https://www.cliffsnotes.com/cliffs-que... 12.1 Assignment (35 pts.) Redesigning Space: The Relationship ... CliffsNoteshttps://www.cliffsnotes.com › ... › Health ScienceCliffsNoteshttps://www.cliffsnotes.com › ... › Health...	0%
4	https://ijhssm.org/issue_dcp/Role%20... The integration of Information and Communication Technology (ICT) in education has revolutionized traditional teaching and learning processes, creating new opportunities for enhancing...	0%
5	https://tasmiyakrishblog.wordpress.co... Your Essential Guide to Full-Stack Development: Concepts and ... WordPress.comhttps://tasmiyakrishblog.wordpress.com › 2025/11/22WordPress.comhttps://tasmiyakrishblog.wordpress...	0%
6	https://www.sciencedirect.com/scien... Aug 1, 2016 · Emergency resource allocation, which mainly deals with how to efGciently allocate emergency resources of rescue agencies to process emerging incidents and reduce...	0%

Content

Minor Project Synopsis Report Disaster Allocation System Project Category: Deep-tech Projexa Team Id- 26E2033 Submitted in partial fulfilment of the requirement of the degree of BACHELORS OF TECHNOLOGY in CSE to K.R Mangalam University By Ananya Sharma (2401010056) Diksha kumari (2401010055) Krishika Sinha (2401010113) Taniya (2401010033) Shikha

(2401010063) Under the supervision of Mr. Nandan Kumar Mishra Department of Computer Science and Engineering School of Engineering and Technology K.R Mangalam University, Gurugram- 122001, India January 2026 DECLARATION We hereby declare that this project report is our original work and has not been submitted elsewhere. The information presented here is true and correct to the best of our knowledge. Shikha (2401010063) Ananya Sharma (2401010056) Diksha kumari (2401010055) Krishika Sinha (2401010113) Taniya (2401010033) INDEX S No Content 1. Abstract 2. Introduction 3. Motivation 4. Literature Review 5. Gap Analysis 6. Problem Statement 7. Objectives 8. Tools/platform Used 9. Methodology 10. Experimental Setup 11. Results And Discussion 12. Conclusion 13. Future Work 14. References ABSTRACT Disaster situations such as floods, earthquakes, fires, and other natural or man-made calamities often result in severe loss of life and property. One of the major challenges during such events is the inefficient and delayed allocation of available resources like food supplies, medical aid, rescue teams, and shelter facilities. Lack of proper coordination and real-time information can lead to unequal distribution, wastage of resources, and delayed response in critical areas. The Disaster Allocation System aims to address these challenges by providing a centralized and intelligent platform for managing disaster-related information and resource distribution. The system is designed to collect, organize, and process disaster data in order to assist authorities in making faster and more informed decisions. By maintaining structured records of disaster incidents, available resources, and allocation details, the system ensures transparency and efficiency in emergency response operations. The project is developed using modern backend technologies, enabling secure handling of data and modular system design. It focuses on scalability, so that the system can be extended in the future to include real-time analytics, geographic mapping, and automated decision support. The proposed solution helps in minimizing response time, avoiding resource duplication, and ensuring that aid reaches the most affected areas promptly. Overall, the Disaster Allocation System contributes to improved disaster management by enhancing coordination, optimizing resource utilization, and supporting timely relief operations, thereby reducing the overall impact of disasters on society. 1. INTRODUCTION Rapid urbanization, climate change, unplanned infrastructure growth, and population expansion have significantly increased the frequency and impact of disasters across the world. Natural disasters such as floods, earthquakes, cyclones, and landslides, along with man-made emergencies like industrial accidents and fires, pose serious threats to human life and economic stability. In developing countries like India, these challenges are further intensified due to inadequate disaster preparedness, limited resources, and poor coordination among relief agencies. One of the major issues observed during disaster situations is the inefficient allocation and management of relief resources. Even when sufficient aid is available, the absence of a structured system often results in delayed response, unequal distribution, and mismanagement of essential supplies such as food, medical assistance, shelter, and rescue services. This lack of coordination can worsen the situation, leading to avoidable losses and prolonged recovery periods. The COVID-19 pandemic further exposed the vulnerabilities of existing emergency response systems. Lockdowns, supply chain disruptions, and sudden surges in demand highlighted the need for a reliable and centralized mechanism to manage emergency resources efficiently. Similar challenges are observed during large-scale disasters, where manual record-keeping and unorganized communication fail to meet the urgency of the situation. In such scenarios, a technology-driven solution becomes essential. A Disaster Allocation System provides a centralized platform that helps authorities monitor disaster events, track available resources, and ensure timely and fair distribution to affected areas. By enabling structured data management and improved coordination, the system plays a crucial role in strengthening disaster response mechanisms and reducing the overall impact of emergencies on society. 2. MOTIVATION In recent years, the world has witnessed a rapid increase in the frequency and severity of disasters due to climate change, uncontrolled urbanization, population growth, and inadequate planning. Natural calamities such as floods, earthquakes, cyclones, and landslides, along with man-made disasters like industrial accidents and fires, have caused massive loss of life and property. In countries like India, these situations are often aggravated by poor coordination among authorities and inefficient distribution of relief resources. Although disaster management authorities make significant efforts to provide aid, the lack of a centralized and well-structured allocation system results in delays, duplication of resources, and unequal distribution of relief. In many cases, some regions receive excess supplies while others suffer due to shortages of essential necessities such as food, medical aid, shelter, and rescue equipment. These inefficiencies highlight the urgent need for a system that can streamline disaster response and ensure optimal utilization of available resources. The COVID-19 pandemic further exposed the shortcomings of traditional emergency management systems. Sudden spikes in demand, logistical challenges, and lack of real-time information made it extremely difficult to manage and allocate resources effectively. Similar challenges are observed during large-scale disaster events, where manual processes and fragmented data lead to poor decision-making under time-sensitive conditions. The motivation behind this project is to design and develop a Disaster Allocation System that enhances the efficiency, transparency, and speed of resource distribution during emergency situations. By providing a centralized platform to manage disaster information, track available resources, and support informed allocation decisions, the system aims to minimize response delays and reduce the overall impact of disasters. This project seeks to contribute toward a more reliable and technology-driven disaster management framework that can support authorities in saving lives and restoring normalcy more effectively. 3. LITERATURE REVIEW DISASTER MANAGEMENT AND RESOURCE ALLOCATION SYSTEMS: Several studies highlight the critical role of efficient resource allocation in disaster management. Research emphasizes that improper planning and delayed distribution of relief materials significantly increase casualties and economic losses during disasters. Traditional disaster response systems largely depend on manual coordination among multiple agencies, which often leads to miscommunication, duplication of efforts, and uneven distribution of essential resources. Studies suggest that centralized disaster management platforms can improve coordination by enabling authorities to monitor disaster severity, available resources, and response status in real time, thereby improving decision-making during emergencies. ICT-BASED DISASTER

RESPONSE SYSTEMS: The integration of Information and Communication Technology (ICT) in disaster management has been widely explored to improve emergency response efficiency. Several researchers propose web-based and cloud- based disaster management systems that allow authorities to collect, store, and analyze disaster-related data. These systems enable faster dissemination of information among stakeholders such as government agencies, NGOs, and rescue teams. However, many existing ICT-based systems focus primarily on data reporting and lack effective mechanisms for automated or optimized resource allocation, which limits their practical effectiveness during large-scale disasters RESOURCE TRACKING AND ALLOCATION USING DATABASE SYSTEMS: Various studies have explored database-driven approaches for managing disaster resources such as food supplies, medical aid, rescue equipment, and shelters. These systems maintain structured records of resources, their locations, and availability status. While such approaches improve transparency and record-keeping, they often rely on static data and manual updates, making them less suitable for rapidly changing disaster scenarios. Researchers emphasize the need for dynamic systems that can update resource availability in real time and support quick allocation decisions based on urgency and demand. DECISION SUPPORT SYSTEMS FOR DISASTER MANAGEMENT; Decision Support Systems (DSS) have been proposed to assist authorities in evaluating disaster severity and prioritizing resource distribution. These systems use predefined rules, historical data, and analytical models to suggest optimal allocation strategies. Studies show that DSS can significantly reduce response time and improve resource utilization. However, many existing systems are complex, expensive to implement, and require extensive training, which restricts their adoption in developing countries. WEB-BASED DISASTER MANAGEMENT PLATFORMS; Recent research highlights the growing adoption of web-based disaster management platforms due to their accessibility and scalability. These platforms allow multiple users to access the system simultaneously, update disaster information, and track relief operations from different locations. While such systems improve collaboration, most lack role- based access control and proper authorization mechanisms, raising concerns regarding data security and misuse. This creates a need for a secure, role-driven disaster allocation system that balances accessibility with data integrity. Table: Literature Review on Disaster Management and Resource Allocation Systems S. No. Title / Area Methodology Used Key Findings Limitations Identified 1 Disaster Management and Resource Allocation Manual coordination and centralized planning Highlighted the importance of timely resource allocation to reduce casualties Manual processes lead to delays and miscommunication 2 ICT-Based Disaster Response Systems Web-based information systems Improved communication between disaster response agencies Lack of automated resource allocation mechanisms 3 Database-Driven Disaster Resource Management Relational databases for resource tracking Enhanced transparency and record maintenance Static data; not suitable for real-time disaster scenarios 4 Decision Support Systems for Disaster Management Rule-based decision support models Assisted authorities in prioritizing resource distribution Complex implementation and high cost 5 Cloud-Based Disaster Management Platforms Cloud storage and online dashboards Enabled remote access and multi-agency collaboration Security and access control concerns 6 Emergency Response Coordination Systems Centralized control systems Reduced response time during emergencies Poor scalability during large-scale disasters 7 Real-Time Disaster Monitoring Systems Sensor and data- driven monitoring Provided faster situational awareness High dependency on infrastructure availability 8 Web-Based Relief Distribution Systems Online portals for relief tracking Improved accountability and tracking of relief materials Limited user role management and authorization 9 Automated Disaster Resource Allocation Models Optimization and scheduling algorithms Improved efficiency in resource utilization Difficult to implement in real-world dynamic conditions 10 Integrated Disaster Management Systems Multi-module system architecture Combined disaster reporting, monitoring, and response Lack of modularity and flexibility 4. GAP ANALYSIS From the extensive research conducted in the field of disaster management, it is evident that technology has been increasingly used to improve disaster response through ICT-based systems, web platforms, database management, and decision support tools. Many existing studies focus on disaster reporting, early warning systems, and post-disaster assessment, which help authorities understand the severity and impact of disasters. While these systems enhance information sharing and documentation, they often function in isolation and lack integrated mechanisms for efficient resource allocation. Most current disaster management solutions emphasize data collection and communication rather than real-time, intelligent allocation of relief resources such as food, medical aid, rescue teams, and shelters. In many cases, resource allocation decisions are still carried out manually, leading to delays, duplication of efforts, unequal distribution, and wastage of critical supplies. Additionally, several decision support systems proposed in research are complex, costly, and difficult to deploy, making them unsuitable for developing countries with limited infrastructure and technical expertise. Furthermore, existing systems rarely provide a centralized platform that combines disaster incident tracking, resource availability monitoring, and allocation records in a structured and transparent manner. The absence of such an integrated approach reduces coordination among authorities and relief agencies, especially during large-scale emergencies where timely response is crucial. The proposed Disaster Allocation System addresses these gaps by offering a centralized, scalable, and structured platform for managing disaster- related data and resource distribution. By focusing on efficient allocation, transparency, and future extensibility, the system aims to overcome the limitations of existing solutions and provide a practical, technology-driven approach to improve disaster response and relief operations. 5. PROBLEM STATEMENT During disaster situations such as floods, earthquakes, cyclones, fires, and pandemics, timely and efficient allocation of relief resources is critical for saving lives and minimizing damage. However, existing disaster management practices often rely on manual coordination, fragmented data, and unstructured communication among multiple agencies. This results in delays in response, unequal distribution of resources, duplication of efforts, and wastage of essential supplies such as food, medical aid, rescue teams, and shelter facilities. In many cases, authorities lack a centralized system to track disaster incidents, monitor real-time availability of resources, and record allocation details. The absence of an integrated and

transparent platform makes it difficult to prioritize affected areas based on severity and urgency, leading to inefficient decision-making during time-sensitive emergencies. These challenges are further intensified in large-scale disasters where rapid response and coordination are crucial. Therefore, the problem lies in designing a centralized and efficient Disaster Allocation System that can systematically manage disaster information, track available resources, and support timely and fair allocation of relief resources. Such a system should improve coordination among authorities, reduce response time, and ensure that aid reaches the most affected regions effectively. 6. OBJECTIVES Sample Objectives 1. To design and develop a centralized Disaster Allocation System that efficiently manages disaster-related information and coordinates the distribution of relief resources during emergency situations. 2. To create a structured database for maintaining real-time records of disaster incidents, available resources such as food, medical aid, rescue teams, and shelters, and their allocation status. 3. To improve decision-making during disasters by enabling authorities to prioritize affected areas based on severity, urgency, and resource availability. 4. To ensure transparency and accountability in the allocation process by maintaining detailed allocation logs and minimizing duplication or misuse of resources. 5. To develop a scalable and secure system architecture that can be extended in the future to support real-time analytics, geographic mapping, and automated decision-support features. The objective of this project is to bridge the gaps identified in existing disaster management systems by providing a unified and efficient platform for disaster resource allocation. The proposed system aims to reduce response time, optimize resource utilization, and enhance coordination among authorities, thereby supporting effective disaster relief operations and minimizing the overall impact of disasters on society. 7. Tools/Technologies Used For the development of the Disaster Allocation System, a combination of modern frontend, backend, and database technologies has been used to ensure efficient data management, user-friendly interaction, system scalability, and reliability during emergency situations. Each technology has been carefully selected based on its suitability for handling disaster- related information and resource allocation processes. FRONTEND TECHNOLOGIES HTML (HyperText Markup Language) HTML is used to design the basic structure of the web application. It provides the layout for different modules such as disaster reporting, resource management, and allocation dashboards. HTML enables the creation of forms, tables, and reports required for systematic data entry and display. CSS (Cascading Style Sheets) CSS is used to enhance the visual appearance of the web application. It helps in designing a responsive, clean, and user-friendly interface by controlling layout, colors, fonts, and spacing. A well-designed interface is essential for quick access to information during disaster situations. JavaScript JavaScript is used to add interactivity and dynamic behavior to the frontend. It enables real-time form validation, dynamic content updates, and smooth user interaction without frequent page reloads. JavaScript improves system responsiveness and enhances the overall user experience. BACKEND TECHNOLOGY The backend of the Disaster Allocation System is responsible for processing user requests, managing disaster and resource data, and enforcing system logic. It handles authentication, authorization, and communication between the frontend and the database. The backend ensures secure, reliable, and efficient data flow, which is critical during emergency response operations. DATABASE MANAGEMENT SYSTEM: MONGODB MongoDB is used as the Database Management System (DBMS) for the Disaster Allocation System. It is a NoSQL, document-oriented database that stores data in JSON-like documents, making it suitable for managing dynamic and large-scale disaster-related data. MongoDB provides high performance, flexibility, and scalability, which are essential for handling rapidly changing information such as disaster incidents, resource availability, and allocation records. Its schema-less structure allows easy modification and future expansion of the system. Additionally, MongoDB supports high availability and fast data retrieval, ensuring uninterrupted system operation during critical situations. REASONS FOR SELECTING THESE TECHNOLOGIES 1. Efficient Data Handling – Supports structured and dynamic disaster data management. 2. Scalability – Allows the system to grow with increasing data and users. 3. User-Friendly Interface – Enables quick access and ease of use during emergencies. 4. Fast Processing and Response – Ensures timely allocation of resources. 5. Secure and Reliable – Protects sensitive disaster and allocation data. 6. Future Extensibility – Can be enhanced with analytics, mapping, and decision support features. 8. METHODOLOGY 1. DISASTER DATA COLLECTION MODULE This module is responsible for registering and monitoring disaster incidents. Information such as disaster type, affected location, severity level, date, and description is collected through the system interface. The collected data is stored in the database in a structured format, allowing authorities to track multiple disaster events simultaneously. This module ensures that disaster information is consistently updated, enabling authorities to analyze the situation accurately and respond promptly. 2. RESOURCE MANAGEMENT MODULE The resource management module maintains detailed records of available relief resources such as food supplies, medical aid, rescue teams, transportation facilities, and shelters. Each resource entry includes information regarding quantity, location, availability status, and last updated time. This module helps in preventing duplication of resources and ensures that authorities have real-time visibility of available aid, which is critical during large-scale disaster situations. 3. RESOURCE ALLOCATION MODULE The resource allocation module is the core component of the system. It processes disaster severity and resource availability data to allocate resources efficiently to affected areas. Allocation decisions are made based on predefined criteria such as urgency, disaster impact, and resource availability. Once resources are allocated, the allocation details are recorded in the database along with timestamps to ensure transparency and accountability. This systematic allocation process reduces delays and ensures fair distribution of relief materials. 4. MONITORING AND TRACKING MODULE This module continuously tracks the status of allocated resources and disaster response activities. It allows authorities to monitor whether allocated resources have been dispatched, delivered, or are still pending. Any updates or changes are logged in the system for future reference. The monitoring feature improves coordination among agencies and helps identify bottlenecks in relief operations. 5. USER MANAGEMENT AND ACCESS CONTROL MODULE The system implements role-based access control to ensure data security and integrity. Different users such as administrators, disaster management officials, and field coordinators are

granted access based on their roles and responsibilities. This module prevents unauthorized access and ensures that sensitive disaster and allocation data is handled securely. 6. REPORTING AND LOGGING MODULE This module generates structured reports related to disaster incidents, resource utilization, and allocation history. Logs are maintained for every critical operation, providing a transparent record of decisions and actions taken during disaster response. These reports assist authorities in evaluating system performance and planning future disaster preparedness strategies. Figure 2: Architectural Diagram This system is designed to manage disaster situations efficiently by processing disaster-related data and allocating resources accordingly. It begins with inputting disaster details such as type, location, severity, and time. The system then processes this data to determine priorities and interacts with a resource database (food, medical supplies, shelters, and teams) and a user management module for secure access. Based on the severity and need, the resource allocation module distributes resources, which are tracked in real-time by the monitoring module. Finally, reports and logs are generated for transparency, analysis, and future planning. The system can be implemented in Python using various modules for data handling, tracking, and reporting. 9.Experimental Setup The Disaster Allocation System was developed and tested. The frontend was built using HTML, CSS, and JavaScript, while the backend was implemented using FastAPI in Python. MongoDB was used as the database for storing disaster, resource, and allocation data. The system was tested by performing various operations such as adding disaster data, managing resources, and allocating resources through the frontend interface. API requests were handled by the backend and the results were stored and retrieved from the MongoDB database. The system also includes role-based access control, where different users such as administrators and volunteers can log in and access functionalities based on their roles. The following screenshots demonstrate the working of the complete system. Figure 1: Frontend interface of the Disaster Allocation System Figure 2: Volunteer login interface of the system. Figure 3: Adding disaster data through frontend interface Figure 4: Map-based visualization showing disaster locations and resource requirements Figure 5: Admin login interface with access control features Figure 6: Resource allocation result based on disaster severity Figure 7: Resource status showing “In Transit” during delivery process 10.RESULTS AND DISCUSSION The Disaster Allocation System was successfully implemented and tested. Results: - Disaster information is stored and retrieved correctly - Resource management module works efficiently - Allocation records are maintained properly - APIs return correct responses during testing Discussion: The system improves coordination and ensures structured handling of data. It reduces manual effort and helps in faster decision-making. Limitations: - Frontend and backend integration is partially completed - No real-time data updates yet 11.CONCLUSION The Disaster Allocation System has been successfully designed and developed to address the challenges faced during disaster situations, particularly in the efficient management and allocation of resources. The system provides a centralized platform for storing disaster-related data, tracking available resources, and maintaining allocation records in a structured manner. By replacing manual processes with a digital system, it helps reduce delays, minimize resource wastage, and improve coordination among authorities. The project ensures transparency and better decision-making by maintaining proper logs and records of all operations. Overall, the system contributes to improving disaster response efficiency and supports timely distribution of resources to affected areas. Although the current implementation focuses mainly on backend functionality, it provides a strong foundation for building a complete and scalable disaster management solution in the future. 12.FUTURE WORK The Disaster Allocation System has been successfully developed with an integrated frontend and backend, providing a complete platform for managing disaster data and resource allocation. However, the system can be further enhanced with advanced features to improve its real-world applicability. In the future, a real-time tracking system can be implemented to monitor the movement and delivery status of allocated resources such as food, medical supplies, and rescue teams. This will ensure that resources reach the affected areas on time and improve transparency in the distribution process. Additionally, machine learning techniques can be incorporated to predict disaster severity and automatically suggest optimal resource allocation strategies based on historical data. These enhancements will make the system more intelligent, scalable, and capable of handling complex real-world disaster management scenarios. 13.REFERENCES • Su, Z., Zhang, G., Liu, Y., Yue, F., & Jiang, J. (2016). Multiple emergency resource allocation for concurrent incidents in natural disasters. International Journal of Disaster Risk Reduction, 16, 145–158. https://doi.org/10.1016/j.ijdrr.2016.01.007 • Altay, N. (2012). Capability-based resource allocation for effective disaster response. IMA Journal of Management Mathematics, 24(2), 253–273. https://doi.org/10.1093/imaman/dpr017 • Rye, S., & Aktas, E. (2022). A multi-attribute decision support system for allocation of humanitarian cluster resources. Sustainability, 14(20), 13423. https://doi.org/10.3390/su142013423 • Zhou, Y., Chen, X., & Huang, Z. (2023). Location–allocation problem in emergency logistics considering lateral transshipment strategy. Computers & Industrial Engineering, 180, 109365. https://doi.org/10.1016/j.cie.2023.109365 • Li, X., Sun, P., & Wang, H. (2025). Strategic allocation of medical resources in disaster response: Multi-objective dynamic optimization. Computers & Industrial Engineering, 190, 108551. https://doi.org/10.1016/j.cie.2025.108551 • Moghaddam, M. H., & Torabi, S. A. (2022). Multi-period equitable and efficient disaster relief allocation under uncertainty. Annals of Operations Research, 320, 531–558. https://doi.org/10.1007/s10479-022-04815-7 • Zhang, Y., & Huang, M. (2013). Resource redistribution under lack of information in disaster scenarios. arXiv preprint arXiv:1312.6339. https://arxiv.org/abs/1312.6339 • Zhao, Q., & Wang, L. (2021). Game-theoretic models for resource allocation during floods. arXiv preprint arXiv:2112.01439. https://arxiv.org/abs/2112.01439